MODEL LAB · 2026-10-17

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

26,271
held-out plate appearances

3.07%
held-out HR rate

0.6059
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-04, scored on the 26,271 that came after (actual HR rate 3.07%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5794	1.65×	0.02968	0.13619	0.00571
KNN comps Prior is the pooled rate of the nearest comparables.	0.6059	1.56×	0.02962	0.13488	0.00390
SVR Support-vector regression on the same feature matrix.	0.5972	1.56×	0.02967	0.13558	0.00596
Random forest Forest regression; a learned similarity metric.	0.5982	1.55×	0.02964	0.13516	0.00523
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6059	1.56×	0.02962	0.13488	0.00390
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6059	1.56×	0.02962	0.13488	0.00390
Linear regression Ordinary least squares on the same features.	0.6017	1.58×	0.02965	0.13518	0.00398
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6027	1.65×	0.02966	0.13524	0.00623
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6007	1.60×	0.02964	0.13512	0.00470
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5988	1.58×	0.02966	0.13527	0.00582

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **251** knn **251** svr **251** rf **251** core **251** ensemble **251** linear **251** logistic **251** nn **251** xgb **251**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,786 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5860	1.36×	0.02314	0.11229	0.01037
KNN comps Prior is the pooled rate of the nearest comparables.	0.6232	1.97×	0.02306	0.10992	0.00977
SVR Support-vector regression on the same feature matrix.	0.6017	1.82×	0.02325	0.11321	0.01716
Random forest Forest regression; a learned similarity metric.	0.6106	2.12×	0.02316	0.11068	0.01542
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6232	1.97×	0.02306	0.10992	0.00977
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6232	1.97×	0.02306	0.10992	0.00977
Linear regression Ordinary least squares on the same features.	0.6211	1.97×	0.02326	0.11135	0.01418
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6307	1.66×	0.02325	0.11095	0.01676
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6236	1.82×	0.02308	0.11000	0.01268
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6196	1.82×	0.02326	0.11125	0.01972

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jonathan Aranda 12.8%, Jo Adell 12.3%, Junior Caminero 12.1%, Patrick Bailey 11.6%, Chase DeLauter 10.6%, Victor Mesa Jr. 10.2%
Flat league	5/6	Jonathan Aranda 12.1%, Jo Adell 11.6%, Junior Caminero 11.5%, Patrick Bailey 11.0%, Chase DeLauter 10.7%, Travis Bazzana 10.1%
KNN comps	6/6	Jonathan Aranda 12.9%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.7%, Chase DeLauter 10.7%, Victor Mesa Jr. 10.3%
SVR	6/6	Junior Caminero 13.1%, Jo Adell 12.5%, Jonathan Aranda 12.4%, Patrick Bailey 11.5%, Victor Mesa Jr. 10.8%, Chase DeLauter 10.4%
Random forest	6/6	Jonathan Aranda 12.7%, Junior Caminero 12.3%, Jo Adell 12.2%, Patrick Bailey 11.2%, Victor Mesa Jr. 10.7%, Chase DeLauter 10.6%
Core (KNN+SVR)	6/6	Jonathan Aranda 12.9%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.7%, Chase DeLauter 10.7%, Victor Mesa Jr. 10.3%
Forest blend	6/6	Jonathan Aranda 12.9%, Jo Adell 12.0%, Junior Caminero 12.0%, Patrick Bailey 11.7%, Chase DeLauter 10.7%, Victor Mesa Jr. 10.3%
Linear regression	6/6	Jonathan Aranda 12.7%, Jo Adell 12.2%, Junior Caminero 12.1%, Patrick Bailey 11.4%, Victor Mesa Jr. 10.5%, Chase DeLauter 10.4%
Logistic regression	6/6	Jo Adell 12.6%, Jonathan Aranda 12.5%, Junior Caminero 12.5%, Patrick Bailey 11.1%, Victor Mesa Jr. 10.5%, Chase DeLauter 10.3%
Neural net (torch)	6/6	Jonathan Aranda 12.6%, Junior Caminero 12.5%, Jo Adell 12.5%, Patrick Bailey 11.3%, Victor Mesa Jr. 10.6%, Chase DeLauter 10.3%
XGBoost	6/6	Junior Caminero 12.9%, Jonathan Aranda 12.7%, Jo Adell 12.4%, Patrick Bailey 11.1%, Victor Mesa Jr. 10.4%, Chase DeLauter 10.4%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jonathan Aranda**	TB slate	**12.8%**	12.1%	12.9%	12.4%	12.7%	12.9%	12.9%	12.7%	12.5%	
2	**Jo Adell**	CLE slate	**12.3%**	11.6%	12.0%	12.5%	12.2%	12.0%	12.0%	12.2%	12.6%	
3	**Junior Caminero**	TB	**12.1%**	11.5%	12.0%	13.1%	12.3%	12.0%	12.0%	12.1%	12.5%	
4	**Patrick Bailey**	CLE	**11.6%**	11.0%	11.7%	11.5%	11.2%	11.7%	11.7%	11.4%	11.1%	
5	**Chase DeLauter**	CLE	**10.6%**	10.7%	10.7%	10.4%	10.6%	10.7%	10.7%	10.4%	10.3%	
6	**Victor Mesa Jr.**	TB	**10.2%**	10.0%	10.3%	10.8%	10.7%	10.3%	10.3%	10.5%	10.5%	
7	**Travis Bazzana**	CLE	**10.2%**	10.1%	10.1%	9.8%	9.9%	10.1%	10.1%	10.0%	9.7%	
8	**Nathaniel Lowe**	CLE	**10.0%**	9.9%	10.2%	10.0%	10.1%	10.2%	10.2%	10.0%	10.0%	
9	**Yandy Díaz**	TB	**9.6%**	9.9%	9.7%	10.1%	9.8%	9.7%	9.7%	9.8%	9.6%	
10	**José Ramírez**	CLE	**9.1%**	9.3%	9.2%	9.3%	9.3%	9.2%	9.2%	9.5%	9.7%	
11	**Petey Halpin**	CLE	**8.3%**	9.2%	8.3%	8.7%	8.5%	8.3%	8.3%	9.3%	8.9%	
12	**Brayan Rocchio**	CLE	**6.3%**	6.8%	6.2%	6.3%	5.9%	6.2%	6.2%	6.0%	6.0%	
13	**Liam Hicks**	TB	**6.1%**	7.0%	6.1%	6.9%	6.0%	6.1%	6.1%	6.5%	6.3%	
14	**Jonny DeLuca**	TB	**5.6%**	6.1%	5.5%	5.5%	5.2%	5.5%	5.5%	5.5%	5.7%	
15	**Richie Palacios**	TB	**5.4%**	6.2%	5.5%	5.3%	5.0%	5.5%	5.5%	5.4%	5.4%	
16	**Steven Kwan**	CLE	**4.4%**	7.4%	4.7%	3.1%	3.6%	4.7%	4.7%	3.0%	4.4%	
17	**Taylor Walls**	TB	**3.6%**	5.1%	3.7%	3.0%	3.2%	3.7%	3.7%	3.8%	3.8%	
18	**Chandler Simpson**	TB	**2.5%**	5.4%	3.1%	1.8%	1.4%	3.1%	3.1%	1.4%	2.8%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-17 · model 2026.10.01-stack · built 2026-10-18 04:22:05 · live data 2026-10-18 09:31:00

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